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1 Peano Arithmetic

If $n \in N$ is nonzero, does there exist $m \in N$ such that $n = m++$? We proceed by induction. Let $P(n)$ denote the proposition that $n = 0$ or there exists $m \in N$ such that $n = m++$. $P(0)$ is true. If $P(n)$ is true, $P(n++)$ is true (regardless of $P(n)$ anyway).

Let $\{f_n\}$ be functions $f_n : N \rightarrow N$ and $c \in N$. Then there exists a unique function $a : N \rightarrow N$ such that $a(n++) = f_n(a(n))$.

(2) Uniqueness. Suppose a, b are such functions. Let $P(n)$ denote the proposition that $a(n) = b(n)$. $P(0)$ is true since $a(0) = b(0) = c$. If $P(n)$ is true, $a(n) = b(n)$. Then $a(n++) = f_n(a(n)) = f_n(b(n)) = b(n++)$, so $P(n++)$ is true.

Define $+_n : N \rightarrow N$ by $+_n(0) = n$, $+_n(m++) = +_n(m)++$, so by before $+_n$ is unique.

Claim $+_0(n) = n$. We proceed by induction. Let $P(n)$ denote the statement $+_0(n) = n$. $+_0(0) = 0$. If $+_0(n) = n$, then $+_0(n++) = +_0(n)++ = n++$.

Claim: $+_a(+_b(c)) = +_{+_a(b)}(c)$. We induct on c . $P(0) : +_a(+_b(0)) = +_a(b) = +_{+_a(b)}(0)$. $P(c) \implies P(c++)$. $+_a(+_b(c++)) = +_a(+_b(c)++) = +_a(+_b(c))++ = +_{+_a(b)}(c)++ = +_{+_a(b)}(c++)$.

Claim: $+_a(b) = +_b(a)$?

Let $a \leq b$ if there exists $d \in N$ such that $b = +_a(d)$. If $a \leq b$ and $b \leq c$, let $c = +_b(e)$. Then $c = +_b(e) = +_{+_a(d)}(e) = +_a(+_d(e))$, so $c \geq a$. If $a \leq b$ and $b \leq a$, then $b = +_a(d)$ and $a = +_b(e)$, so $a = +_a(+_d(e))$ (so $\leq$ is transitive).

Claim: $+_{a++}(b) = +_a(b++)$. We induct on b . $+_{a++}(0) = a++ = +_a(0)++ = +_a(0++)$. $+_{a++}(b++) = +_{a++}(b)++ = (+_a(b++))++ = +_a((b++)++)$.

Claim: $+_a(d) = a \implies d = 0$. We induct on a . $P(0)$. $+_0(d) = d = 0$.

$P(a) \implies P(a+1)$. $a++ = +_{a++}(d) = +_a(d++) = +_a(d)++ \rightarrow +_a(d) = a \rightarrow d = 0$.

If $a = +_a(+_d(e)) \implies +_d(e) = 0$. If $e \neq 0$, then $e = e'++$, so $+_d(e) = +_d(e'++) = +_d(e')++ \neq 0$. Then $e = 0$, so $+_d(e) = d = 0$. From $b = +_a(d) = +_a(0) = a$. Hence $\leq$ is antisymmetric.

Claim: $+_a(b) = +_b(a)$. We induct on a . $P(0)$. $+_0(b) = b = +_b(0)$. $P(a) \implies P(a++)$. $+_{a++}(b) = +_a(b++) = +_a(b)++ = +_b(a)++ = +_b(a++)$.

Claim: if $a+b = a+c$, then $b = c$. We induct on a . $P(0)$. $0+b = 0+c \implies b = c$. $P(a) \implies P(a++)$. $(a++)+b = (a++)+c \implies (a+b)++ = (a+c)++ \implies a+b = a+c \implies b = c$.

Claim: if $a+c = b+c \implies a = b$. We induct on c . $P(0)$. $a+0 = b+0 \implies a = b$. $P(c) \implies P(c++)$. $a+(c++) = b+(c++) \implies (a+c)++ = (b+c)++ \implies a+c = b+c \implies a = b$.

2 Order of the Naturals

Claim: addition preserves order. If $a \geq b$, then $a+c \geq b+c$. $a = b+d$, so $a+c = (b+d)+c = b+(d+c) = b+(c+d) = (b+c)+d$.

Claim: addition preserves order (other direction). $a \geq b \implies c+a \geq c+b$. $c+a = c+(b+d) = (c+b)+d$.

Claim: $a < b$ iff $a++ \leq b$. If $a < b$, then $b = a+d$ for $d \neq 0$. Then $d = e++$, so $b = a+(e++) = (a++)+e$. If $a++ \leq b$, then $b = (a++)+e = a+(e++)$, so $b \geq a$. If $b = a$, then $e++ = 0$, cont.

Claim: if $a, b \in N$, then $a < b, a = b, a > b$, where the cases are mutually exclusive. We induct on a . $P(0)$. If b is positive, then $b > 0$. Else, $b = 0$. Sps FTSOC $b < 0$, so $0 = b+d \implies b = d = 0$, cont. $P(a) \implies P(a++)$. We have $a < b, a = b, a > b$. If $a < b$, then $a++ \leq b$, so $a++ < b$ or $a++ = b$. If $a = b$, then $a++ > b$. If $a > b$, then $a++ > a > b$.

Claim (Well-Ordering of N): Let $\emptyset \subsetneq S \subset N$. Then S has a minimal element. Sps FTSOC S doesn't have a minimal element. Let $a_0 \in S$. (AC?) Pick $a_{n++} < a_n$, both in S , so we have $a : N \rightarrow N$ such that $a_{n++}++ \leq a_n$. Claim: $a_{n+d}+d \leq a_n$. $a_n \leq a_n$. $a_{n+(d++)}+(d++) = (a_{(n+d)++}++) + d \leq a_{n+d}+d \leq a_n$. Take $n = 0, d = a_0++$. Then $a_0++ \leq a_d + a_0++ \leq a_0$, cont.

Strong Induction. Let $n_0 \in N$ and let P be a proposition such that $P(n_0)$

is true and for all $n \geq n_0$, $P(k)$ for all $k < n$ implies $P(n)$. Then $P(n)$ is true for all $n \geq n_0$.

Let $S \subset N$ denote the set of $n \geq n_0$ such that $P(n)$ doesn't hold. Let $m \in S$ be minimal. Note $m \neq n_0$. Else, $m > n_0$ and for all $n < m$ we have $P(n)$ true, so $P(m)$ is true, cont.

Let $Q(n) = (\forall m_0 \leq m' < m_0 + n, P(m'))$. $Q(0) = (\forall m_0 \leq m' < m_0, P(m')) = T$. Suppose $Q(n)$ holds. Let $m_0 \leq m' < m_0 + (n++)$, so $m_0 \leq m' \leq m_0 + n$. For $m_0 \leq m' < m_0 + n$, $P(m')$ holds. $m' = m_0 + n$ holds by IH, so $Q(n++)$ holds.

3 Multiplication

If $n, m \neq 0$, then $nm \neq 0$. Let $m = m'++$. Then $nm = n(m'++) = nm' + n \geq n > 0$.

Multiplication preserves order. If $a < b$ and $c > 0$, then $ac < bc$. $a \cdot 1 < b \cdot 1$. If $ac < bc$, then $ac + a < bc + b \implies a(c++) < b(c++)$. Tao's proof: let $b = a + d$ for $d > 0$, so $bc = (a + d)c = ac + dc > ac$.

Division algorithm. Let $n, q \in N$ and $q \neq 0$. Then there exist unique m, r fw $n = mq + r$. Let $S = \{k \in N : kq \leq n\}$. Note that $k \leq n$ because if $k \geq n + 1$, then $kq \geq (n + 1)1 > n$, so S is bounded. Sps FTSOC S has no maximal element ($S \neq \emptyset$ since $0 \in S$). Then we may define inductively $a_n : N \rightarrow N$ increasing, for which $a_n \geq n$, cont max.

Let $m \in S$ be maximal. Since $mq \leq n$ there exists r such that $n = mq + r$. We can show $0 \leq r < q$.